

KATERINA NIKIFOROVA

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EDUCATION

Carnegie Mellon University

BSc Computer Science, 3.7/4.0 GPA

Pittsburgh, PA

May 2025

- Teaching Assistant: Algorithm Design and Analysis (January 2024 - present)
- Relevant Coursework: Computer Graphics, Learning for 3D Vision, Geometry-Based Methods in Vision, Compiler Design, Parallel Programming, Machine Learning

Marathon High School: Valedictorian, 4.0/4.0 GPA

Spring 2021

WORK EXPERIENCE

PEAK6 Capital Management

Chicago

Quantitative Research Intern

June 2024 - August 2024

- Designed and integrated market data pipeline observability using OpenTelemetry and Jaeger code instrumentation
- Trained and quantitatively compared deep RNN, LSTM, and factorization model architectures for effectiveness in embedding time-series data using company stock market data and TensorFlow

Boston Dynamics AI Institute

Boston

R&D Systems Intern

June 2023 - August 2023

- Created an asynchronous Python scheduler to simultaneously direct four Spot robots to perform a sequence of interactive tasks over a shared environment with an 89% completion rate under high-stress, dynamic demo runs
- Designed and developed mapping system for mobile robot rearrangement of environment under human-guidance with high-level planning to correct for object misplacements

IBM

Boston

Security Systems Development Intern

May 2022 - August 2022

- Reimplemented authentication request process for backend refactor of major API security software product in C++
- Developed a configuration parser and API mocking system using Google Mock to enable efficient and flexible end-to-end testing leveraged by team over entire effort

TECHNOLOGY SKILLS

Languages: C++, Python, C, OCaml, Java

Development tools: MongoDB, MySQL, Node.js, PHP, HTML/CSS, OpenCV, OpenTelemetry, Google Test/Mock, Vulkan

System tools: ROS1, ROS2, Linux, Docker, CMake, CUDA, OpenMP, MPI

Data tools: PyTorch, TensorFlow, MATLAB, Pytorch3D

NOTABLE PROJECTS

- Partner-developed and optimized an SSA-based C compiler in OCaml with >90% execution speed of -O1 GCC and a novel automatic copy-collected garbage collection memory system (January 2024 - May 2024)
- Augmented open-source SLAM system (ORB_SLAM) with robustness to dynamic points, reducing map reconstruction error by 9.4% over TUM dataset videos where the system did not previously diverge and preventing map divergence for largely dynamic scenes

RESEARCH

Drone Search and Rescue Effort, AirLab, CMU

Research Assistant

August 2024 - present

- Explored dense, language-aligned 3D semantic scene representations for open-vocabulary navigation queries
- Evaluated projective mapping strategies using recent 2D language-aligned pixel labelling systems (e.g. PixelCLIP, SCLIP)
- Modified CLIP's attention and transformer architecture to automate pixel feature labelling using SAM2, achieving an mIoU score of 21.4%, beating the evaluated performance of SCLIP and PixelCLIP

Autonomous Underwater Vehicle Team, CMU

Robotic Perception Lead and Co-Founder

August 2021 - May 2024

- Finalists every year since founding (2022, 2023, 2024), and 3rd place overall in 2022
- Developed global mapping and tracking system for underwater object awareness using EKF SLAM to globally interpolate relative object positions from array of multi-modal perception sensors
- Created object detection system by integrating geometric computer vision techniques and training YOLOv3 CNN model
- Implemented drivers for camera and sonar system with signal processing for noise reduction of sonar echoes
- Integrated perception stack into autonomy stack using Python, C++, and ROS by writing navigation tasks scoring 1/3 of total points during competition